National SAFE KIDS Campaign®

Follow the Leader

A National Study of Safety Role Modeling among Parents and Children

INTRODUCTION

When it comes to safety behaviors, both common sense and empirical evidence support the notion that children imitate their parents.

Previous studies indicate that parents can serve as both positive and negative role models. Four out of 10 children who ride with unbelted drivers are completely unrestrained[1], and nine out of 10 children who don't wear bike helmets have parents who never wear them.[2] Nine out of 10 children who boat with an adult wearing a personal flotation device also wear one.[3] In another study, 90 to 100 percent of parents reported teaching their children to look both ways when crossing a street, and 80 percent reported teaching their children the meaning of the walk/don't walk symbols.[4] Children clearly look to their parents for guidance on safety rules. For instance, 53 percent of children identified "parents having a rule" as essential to convincing them to wear helmets.[5] Collectively, these findings indicate that a parent's actions and words are both important in teaching children about injury prevention.

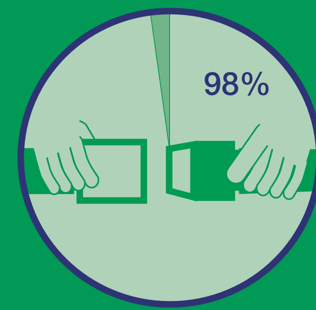
Over the past two decades, the United States has embraced a "culture of safety" in an unprecedented manner. Awareness of childhood injuries as a leading killer and disabler of children has risen. The distribution of safety devices such as child safety seats, bicycle helmets, and smoke detectors has been widespread, and the passage, strengthening and enforcement of safety laws have skyrocketed. Despite a 43 percent decline in the unintentional childhood injury death rate over the past 15 years[6], however, injury remains the leading cause of death among children ages 14 and under in the United States.[7] In 2002, 5,305 children ages 14 and under died and an estimated 115,000 children were totally or partially disabled by their injuries[8], and each year one out of every four children (a total of more than 14 million children ages 14 and under) sustains an injury serious enough to require medical attention.[9, 10]

One reason why children continue to be afflicted by unintentional injury may be that there is a gap between parents' intentions in setting safety rules and their ability to exact compliance with those rules from their children. Using social learning theory[11] as a framework, in this study SAFE KIDS explores the concept that both observation of a behavior and the perceived consequence of that behavior affect a child's imitation of the behavior.[12]

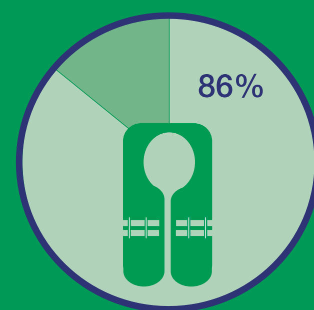
The report focuses on three of the leading causes of injury-related child mortality (motor vehicle occupant injury, pedestrian injury and drowning) and one of the leading causes of child morbidity (bicycle injury). Through a national poll of parents and their children about safety-related knowledge, attitudes and behavior, coupled with observational research of pedestrian safety practices at the community level, SAFE KIDS aims to improve the understanding of safety role modeling in families.

Percentage of parents rating use of safety device as "very important":

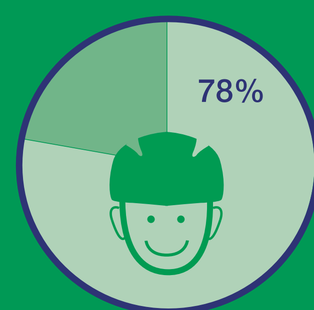
Seat belts



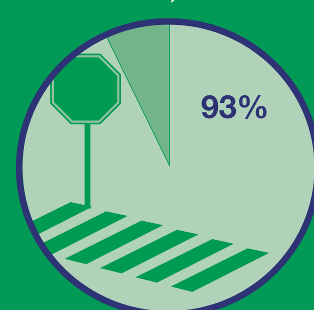
Life jackets



Bike helmets



Crosswalks, if available



METHODOLOGY

Parent and Child Survey Data

To learn more about perceptions and the practice of positive safety behaviors regarding child passenger, water, pedestrian and bicycle safety, SAFE KIDS commissioned Harris Interactive to conduct an online survey of adults and children from December 2004 to January 2005. The results are based on responses from a nationally representative sample of 1,221 U.S. parents of children ages 8 to 14. One child between the ages of 8 and 14 from each parent's household was randomly selected to complete the survey along with the parent. Individual interviews averaged 16 minutes in length. Results were weighted to be representative of U.S. parents of children ages 8 to 14 on key demographic variables such as age, gender, household income, race/ethnicity, education and geographic region.

Pedestrian Observation Data

Data were collected by eight SAFE KIDS coalitions representing eight U.S. states and four regions. A total of 859 groups with an adult accompanying one or more children (859 adults and 1,416 children ages 14 and under) were observed, using instruments and protocols developed by the National SAFE KIDS Campaign.

Each SAFE KIDS coalition selected a minimum of three observation sites, based on volume of adult-child pedestrian groups and the presence of intersections. At least one site observed had a traffic signal, and all had either a traffic signal or a stop sign influencing traffic perpendicular to the crossing.

Each site was observed for 45 minutes by a minimum of two observers. In order to produce a snapshot of role-modeling behavior of adults crossing the street with children, observers recorded the frequency with which adults complied with three common pedestrian safety practices: crossing at the crosswalk, crossing at the appropriate time and visually scanning for approaching traffic.

All coalitions submitted their surveys to the National SAFE KIDS Campaign for analysis. Teleform 7.0 software was used for data entry. Analysis was conducted using SPSS 12.0.

SUMMARY OF RESULTS

The majority of children do look to their parents as safety role models. Eighty-three percent of children consider their mothers safety role models, while 72 percent of children call their fathers safety role models. In addition, 79 percent of children report that other people also have been their safety role models. Sixty-four percent of those children mention their grandparents as role models, while 56 percent name their teachers, 30 percent include their siblings and 26 percent cite their friends.

Children also see themselves as safety role models. Seventy-eight percent report that they have shown other people how to be safe. Sixty-six percent of those children say they are role models for friends, while 47 percent report being role models for siblings.

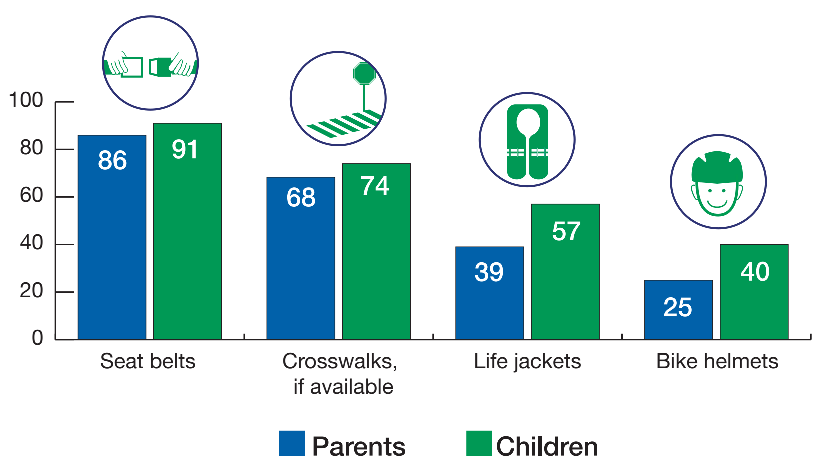
Nearly all parents (98 percent) agree it is important for them to be safety role models for their children, and parents and children concur on the importance of engaging in safe behaviors. Wearing seat belts is seen as the most important safety behavior by both parents and children. Wearing bike helmets and life jackets and crossing at a crosswalk are also viewed as important safety behaviors, but they are practiced less often.

Although parents are more likely to use seat belts than bike helmets, life jackets or crosswalks, they are similarly aware of the risks associated with those areas of safety. Parents show high levels of concern about their children being injured or killed when riding in a car (70 percent), crossing a busy street (66 percent), riding a bike (60 percent) and riding on a boat or being near water (49 percent).

Most parents report that their behaviors became safer when they had children of their own. Seventy-one percent of parents became more likely to wear seat belts, while 42 percent wore bike helmets more often and 47 percent put on life jackets. Sixty-three percent were more likely to use a crosswalk after becoming parents.

Role modeling of safety behavior does not always flow exclusively from parent to child. Both parents and children report that they are more likely to engage in safe behaviors when they are together than when they are not. For example, 72 percent of parents are less likely to cross in the middle of a block when they are with their children. Often children remind their parents to practice safe behaviors. For example, 31 percent of children surveyed had reminded their parents to wear a seat belt when riding in a car.

Percentage reporting that they "always use":



Children who have parents that tend to use safety devices appear more likely to use safety devices themselves.

Child Passenger Safety

Nearly all parents and children ride together in cars often. Fortunately, parents are emphasizing the importance of buckling up, and 91 percent of children say that they always wear a seat belt when riding in a car.

- Almost all parents surveyed (96 percent) believe their children are more likely to wear seat belts if they see their parents wearing them.
- Mothers are slightly more likely than fathers (89 percent versus 82 percent) to stress the importance of using seat belts.
- Seventy-six percent of children surveyed agree they are more likely to wear seat belts if they see their mothers wearing them. Younger children (ages 8 to 10) are more likely than older children (ages 11 to 14) to follow Mom's example (81 percent versus 71 percent), while African American children are more likely to buckle up like Mom than children of Hispanic or other racial backgrounds (86 percent versus 74 percent).

Three factors that influence children to use seat belts, identified by both parents and children, are:

- How important the parent believes seat belt use is. Nearly all parents surveyed (98 percent) believe it is important.
- Whether the parent wears a seat belt when riding in a car. Eighty-six percent of parents report always wearing a seat belt when riding in a car.
- How much the parent worries about his or her children being killed or injured when riding in a car. Seventy percent of parents worry about this possibility.

The high level of seat belt use among parents (86 percent) is matched by widespread use of seat belts among children ages 8 to 14, who have generally outgrown booster seats. Ninety-one percent of children in this age group report that they always wear seat belts when riding in cars.

Bicycle Safety

Sixty percent of parents report riding a bike at least sometimes, while about eight in ten children (82 percent) say they ride bikes at least sometimes. Both parents and children seem to let down their guard about safety during this common activity, especially once children become older.

- Nearly nine in ten parents surveyed (86 percent) believe their children are more likely to wear bike helmets if they see their parents wearing them.
- Parents of younger children are more likely than parents of older children to have talked to their children about wearing a helmet when riding a bike (86 percent versus 72 percent). Younger children are more likely than older children to "usually" or "always" wear a helmet when riding a bike (68 percent versus 51 percent).
- Parents of girls are more likely than parents of boys to say it is very important for their child to always wear a helmet when riding a bike (82 percent versus 74 percent).

Three factors that influence children to use bike helmets, identified by both parents and children, are:

- How important the parent believes helmet use is. Seventy-eight percent of parents surveyed believe it is important.
- Whether the parent uses a helmet when riding a bike. Only 25 percent of parents report always wearing a helmet when riding a bike.
- If the parent talked to his or her child about wearing a helmet. Seventy-nine percent of parents report discussing the topic within the past year.

Although the rate of bike helmet usage is not nearly as high as that for seat belts, 40 percent of children who ride bicycles say they always wear a helmet when riding a bike. This outpaces the 25-percent helmet usage rate of their parents.

Pedestrian Safety

The majority of parents and children say they use safe behaviors when crossing busy streets.

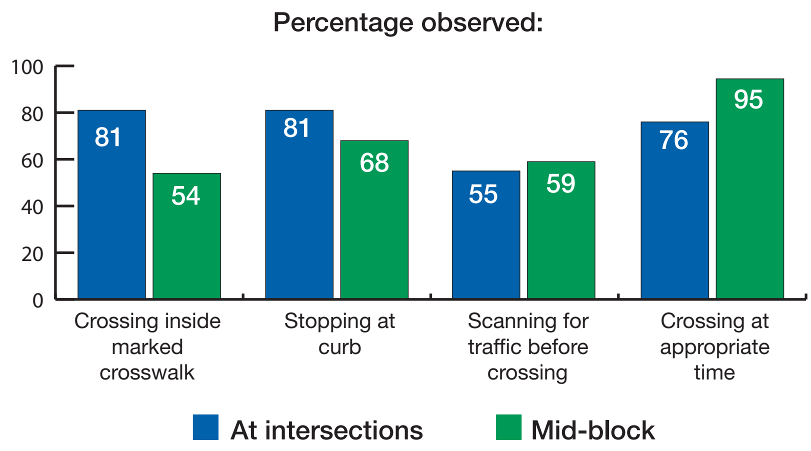
- Almost all parents surveyed (95 percent) believe their children are more likely to cross busy streets safely if they see their parents doing so.
- Mothers are more likely than fathers to always cross at an intersection (68 percent versus 59 percent) or a crosswalk (72 percent versus 61 percent).
- Younger children are more likely than older children to always cross busy streets at an intersection (77 percent versus 67 percent) or a crosswalk (82 percent versus 67 percent).

Four factors that influence children to follow safe pedestrian behaviors, identified by both parents and children, are:

- How important the parent believes using a crosswalk is. Seventy-seven percent of parents surveyed believe it is important.
- How important it is to the parent that his or her child uses a crosswalk, if available, to cross a busy street. Ninety-three percent of parents surveyed believe it is important their children use crosswalks.
- Whether the parent talked to his or her child about using crosswalks. Seventy percent of parents report discussing the topic within the past year.
- How much the parent worries about his or her child being hit by a car when crossing a street. Sixty-six percent of parents worry about this possibility.

Again, children were more likely to report safe pedestrian behavior than their parents, with 68 percent of parents and 74 percent of children saying they always use a crosswalk, if available, when crossing a busy street.

Pedestrian Observation Survey



Most of the parents and other adult caregivers accompanied by one or more child who were observed by SAFE KIDS coalitions crossed busy streets inside marked crosswalks at intersections.

- 81 percent crossed inside a marked crosswalk.
- 10 percent crossed near but outside of a marked crosswalk.

When crossing at mid-block, most parents and caregivers with one or more child crossed inside a marked crosswalk.

- 54 percent crossed inside a marked crosswalk.
- 36 percent crossed near but outside of a marked crosswalk.

Most parents and caregivers with one or more child stopped at the curb before crossing.

- 81 percent stopped at the curb before crossing at an intersection.
- 68 percent were seen stopping at the curb before crossing mid-block.

Before crossing, most parents and caregivers with one or more child scanned for traffic in both directions.

- 55 percent scanned for traffic in both directions before crossing at an intersection.
- 59 percent were observed scanning for traffic in both directions before crossing at a designated area mid-block.

While crossing, some parents and caregivers with one or more child scanned for traffic in both directions.

- 26 percent scanned for traffic in both directions while crossing at an intersection.
- 35 percent were observed scanning for traffic in both directions while crossing at a designated area mid-block.

Nearly all parents and caregivers with one or more child crossed at an appropriate time.

- 76 percent crossed as indicated by a pedestrian signal at an intersection.
- 95 percent were seen crossing at an appropriate time when crossing at a designated area mid-block.

Water Safety

While boating is engaged in somewhat less frequently by parents and children and drowning is caused by a multifaceted cluster of risks, drowning remains the second leading cause of injury-related death among children ages 1 to 14.[13]

- More than nine in ten parents surveyed (92 percent) believe their children are more likely to wear life jackets (also known as personal flotation devices or PFDs) if they see their parents wearing them.
- Parents of younger children are more likely than parents of older children to say it is very important for their child to always wear a life jacket when on a boat (90 percent versus 83 percent). Younger children are more likely than older children to cite the importance of wearing a life jacket when on a boat (80 percent versus 66 percent).

Four factors that influence children to use life jackets, identified by both parents and children, are:

- How important the parent believes life jacket use when on a boat or near water is. Nearly nine in ten parents surveyed (86 percent) believe it is important.
- Whether the parent wears a life jacket when on a boat or near water. 39 percent of parents surveyed report always wearing a life jacket on a boat or near water.
- Whether the parent has talked to his or her child about wearing a life jacket. Sixty-eight percent of parents surveyed report discussing the topic within the past year.
- How much the parent worries about his or her child drowning or suffering a near-drowning incident. Forty-nine percent of parents worry about this possibility.

Children are also more likely to report safe behavior on a boat or near water than their parents. Fifty-seven percent of children say they always wear a life jacket on a boat or near water, as opposed to the 39 percent of parents who report always wearing a life jacket.

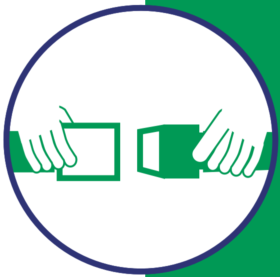
Conclusions

- If the parent perceives a safety behavior to be important, then his or her child appears more likely to practice that behavior.
- Children who have parents who tend to use safety devices appear more likely to use safety devices themselves.
- Both the perception of the importance of seat belt use and actual seat belt use are nearly universal. Safe behavior when in a car is widespread, with 86 percent of parents and 91 percent of children reporting that they always wear seat belts when riding in cars.
- Both parents and children seem to let down their guard about bike helmet safety, especially older children and their parents. Parent and child bike helmet use is substantially lower than other safety behaviors measured in this survey. Only 25 percent of parents and 40 percent of children report that they always wear a helmet when riding a bike.
- Parents and children are more likely to engage in safe behaviors when they are together.
- Nearly all children acknowledge that their parents are safety role models, and parents agree that their role modeling is important.
- Children regard themselves as role models as well.
- In all injury risk areas studied, children are more likely to report safe behavior than their parents are.

SAFE KIDS Recommendations

Parents must learn the essentials about preventing unintentional childhood injuries so that they can teach their children safe behavior. The key actions parents and children must take are:

Child Passenger Safety

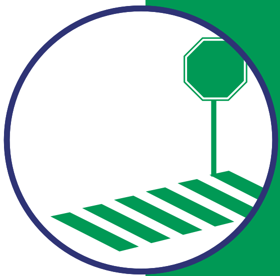


- All children ages 12 and under should be properly restrained in the back seat on every ride.
- Choose and use correctly the right restraint for your child. Any child safety seat must be installed and used according to the manufacturer's instructions and the vehicle owner's manual.
- Obey all traffic laws, including those for child restraint use.
- Parents need to buckle up for every ride, too.

Bicycle Safety

- Every time you ride, wear a bicycle helmet that meets or exceeds the safety standards developed by the U.S. Consumer Product Safety Commission.
- A helmet should sit on top of your head in a level position, and it should not rock forward and backward or side to side. The helmet straps must always be buckled. Bicycle fit and maintenance are also important.
- Learn and follow the rules of the road. Riders should be restricted to sidewalks and paths until they reach age 10 and can show how they follow the rules.
- Obey all traffic laws and teach them to your children, including those for helmet use.

Pedestrian Safety



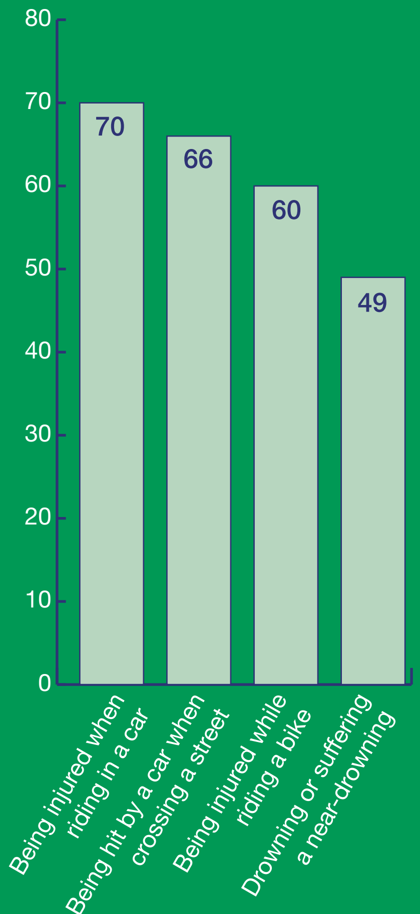
- Teach children to cross streets safely, walk on sidewalks or paths whenever possible and be cautious around cars by demonstrating these behaviors.
- Never allow children under age 10 to cross streets alone.
- Teach children to walk on direct routes with the fewest street crossings.
- Teach your child never to run out into a street for a ball, a pet or any other reason.
- Make sure your child plays in safe places, such as yards, parks and playgrounds – never in the street.

Water Safety

- Always wear a U.S. Coast Guard-approved PFD when on a boat, near an open body of water or when participating in water sports. Air-filled swimming aids, such as "water wings," are not safety devices and cannot substitute for a PFD.
- Always supervise children in and around water. Don't leave, even for a moment. Keep rescue equipment (like a lifesaving ring), a telephone and emergency phone numbers poolside.
- Swim only in designated and supervised swimming areas, and teach children never to swim alone.

- Teach kids the safe way to help someone in trouble in the water: call for help and throw the person something that floats.
- Use barriers to keep kids away from water when you're not around.
- Make sure everyone knows the water safety rules. SAFE KIDS recommends that children take swimming lessons when they're ready, usually after age 4.
- Don't let children dive into water less than nine feet deep, and no one should dive into a river, lake or ocean.

Percentage of parents worried about their child:



Call to Action

To help kids stay safe now, and to teach them how to keep themselves and others safe throughout their lives, the National SAFE KIDS Campaign, its more than 600 coalitions and chapters, and Founding Sponsor Johnson & Johnson will launch a nationwide initiative — Follow the Leader, Safety Starts with You! — during National SAFE KIDS Week 2005 (April 30-May 7).

SAFE KIDS coalitions across the country will participate in SAFE KIDS Week by hosting thousands of family safety fairs and other community events, where parents, caregivers and kids can learn about the importance of adults demonstrating safe behavior. Interactive games and displays will show how to model safe behavior as a pedestrian, while riding a bike or in a motor vehicle, and in or around water.

During National SAFE KIDS Week and throughout the year, SAFE KIDS coalitions join many other public health education and advocacy organizations in a multifaceted approach to reducing unintentional childhood injury, by:

- Advocating for stronger and better enforced child safety laws and regulations.
- Distributing free or low-cost bicycle helmets, car seats and PFDs to children in low-income families.
- Educating families about the importance of using safety devices like these correctly and all the time.
- Encouraging children to take swimming lessons and bicycle riding classes.
- Teaching parents and caregivers to actively supervise children.
- Working with urban planners and engineers to build and maintain safer environments for children.

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Endnotes

[1] Cody BE, Mickalide AD, Paul HP, Colella JM. Child passengers at risk in America: A national study of restraint use. Washington (DC): National SAFE KIDS Campaign, February 2002.

[2] Ehrlich PF, Helmkamp JC, Williams JM, Haque A, Furbee PM. Matched analysis of parent's and children's attitudes and practices towards motor vehicle and bicycle safety: an important information gap. Inj Control Saf Promot. 2004 Mar;11(1):23-8.

[3] Quan L, Bennett E, Cummings P, Trusty MN, Treser CD. Are life vests worn? A multiregional observational study of personal flotation device use in small boats. Inj Prev. 1998 Sep;4(3):203-5.

[4] MacGregor C, Smiley A, Dunk W. Identifying gaps in child pedestrian safety: Comparing what children do with what parents teach. Transportation Research Record 1674. Paper No. 99-0724, 1999.

[5] Cody BE, O'Toole ML, Mickalide AD, Paul HP. A national study of traumatic brain injury and wheel-related sports. Washington (DC): National SAFE KIDS Campaign, May 2002.

[6] National Center for Health Statistics. Centers for Disease Control and Prevention. National Vital Statistics System. 2002 mortality statistics. Hyattsville (MD): National Center for Health Statistics, 2005.

[7] National Center for Health Statistics. Centers for Disease Control and Prevention. National Vital Statistics System. 2002 mortality statistics. Hyattsville (MD): National Center for Health Statistics, 2005.

[8] National Center for Health Statistics. Centers for Disease Control and Prevention. National Vital Statistics System. 2002 mortality statistics. Hyattsville (MD): National Center for Health Statistics, 2005.

[9] Sheppard M. Personal communication. Landover (MD): Children's Safety Network, Economics and Insurance Resource Center, 2005.

[10] Kogan MD, et al. Medically attended nonfatal injuries among preschool-age children: national estimates. American Journal of Preventive Medicine 1995;11(2):99-104

[11] Bandura A. Grusec JE, Menlove FL. Observational learning as a function of symbolization and incentive set. Child Dev 1966 Sep; 37(3):499-506.

[12] Bandura A. Grusec JE, Menlove FL. Observational learning as a function of symbolization and incentive set. Child Dev 1966 Sep; 37(3):499-506.

[13] National Center for Health Statistics. Centers for Disease Control and Prevention. National Vital Statistics System. 2002 mortality statistics. Hyattsville (MD): National Center for Health Statistics, 2005.

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Martin Eichelberger, M.D.
President and CEO

National SAFE KIDS Campaign
1301 Pennsylvania Avenue, NW
Suite 1000
Washington, DC 20004

www.safekids.org

tel 202-662-0600
fax 202-393-2072